

Worker Satisfaction Survey — Data De-identification Workbook

Data De-identification Advisor

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Introduction

This workbook guides you through de-identifying the **Apple Grower Satisfaction Survey** (`WorkerSatisfaction_300rows`, 300 rows, 16 columns).

Learning objectives

- Identify privacy risks in survey data
- Apply de-identification steps in R using **dplyr**
- Export a shareable dataset and a separate data key file

Workflow

1. **Setup** — load packages and set file paths
2. **Load and assess** — inspect raw data and review privacy risks

3. **Transform** — apply de-identification rules step by step
4. **Verify and deliver** — run QA checks and export output files

Output files

- **WorkerSatisfaction_300rows_deidentified.xlsx** — for research team and analysts
- **data_key_file/WorkerSatisfaction_data_key_DUMMY.xlsx** — for authorized personnel only; store separately from the de-identified data (dummy example)

Do not edit the raw source file (**WorkerSatisfaction_300rows.xlsx**).

Roadmap

Part	Section	What happens
1	Setup	Install packages, set paths
2	Load and assess	Read raw data; review direct IDs, dates, geography, free text
3	Transform	Remove, generalize, suppress, and pseudonymize — preview after each step
4	Verify and deliver	QA checks → export de-identified file → export data key

Risk	What to look for (Part 2)	What to do (Part 3)
Direct identifiers	Names, emails, handles, owner names	Remove columns
Dates	age is date of birth	Replace with 5-year age bands
Geography	Small city counts	Suppress rare cities, then anonymize
External sources	Free-text comments	Remove column
Quasi-identifiers	Orchard names	Pseudonymize to Orchard_01 , ...

Part 1 — Setup

Required packages

Package	Purpose
readxl	Read the original Excel survey file
dplyr	Transform data (native pipe <code> ></code>)
writexl	Export Excel outputs

Run **Install packages** once if a package is missing.

```
# install.packages("readxl")
# install.packages("dplyr")
# install.packages("writexl")
# install.packages(c("readxl", "dplyr", "writexl"))
```

```
library(readxl)
library(dplyr)
library(writexl)
```

```
DATA_FILE <- "WorkerSatisfaction_300rows.xlsx"
OUTPUT_FILE <- "WorkerSatisfaction_300rows_deidentified.xlsx"
KEY_DIR <- "data_key_file"
KEY_FILE <- file.path(KEY_DIR, "WorkerSatisfaction_data_key_DUMMY.xlsx")
K_THRESHOLD <- 5
```

k-anonymity threshold: categories with fewer than 5 records will be pooled before anonymization.

```
cat("K_THRESHOLD =", K_THRESHOLD, "\n")
```

```
## K_THRESHOLD = 5
```

```
cat("Data key file:", KEY_FILE, "\n")
```

```
## Data key file: data_key_file/WorkerSatisfaction_data_key_DUMMY.xlsx
```

Part 2 — Load and Assess

Load the unmodified survey and review what makes the data identifiable.

2.1 Load raw data

```
raw <- read_excel(DATA_FILE, sheet = "Unmodified data")
```

```
cat("Raw data:", nrow(raw), "rows ×", ncol(raw), "columns\n")
```

```
## Raw data: 300 rows × 16 columns
```

```
head(raw, 10)
```

```
## # A tibble: 10 × 16
```

```
##   worker_id      email_id age      immigration_stat city province
##   <chr>         <chr>   <dtm>         <chr>         <chr> <chr>
## 1 James Strange james.s~ 1998-09-10 00:00:00 Non-immigrant Kelo~ B.C.
## 2 Maya Liya     maya.li~ 1994-08-04 00:00:00 Immigrant      Kelo~ B.C.
## 3 Amelio Beal   amelio.~ 1995-01-02 00:00:00 Non-permanent r~ West~ B.C.
## 4 Cara Sahara   cara.sa~ 1991-03-22 00:00:00 Non-permanent r~ West~ B.C.
## 5 Neiv Rieg      neivr@~ 1990-07-15 00:00:00 Non-permanent r~ Vict~ B.C.
## 6 Troy Ahoy      tahoy@~ 1988-06-06 00:00:00 Immigrant      Vict~ B.C.
## 7 Dave Mahew     dmahew@~ 2000-11-11 00:00:00 Non-immigrant Vern~ B.C.
## 8 Jamie Thomas   jamiet@~ 1992-12-18 00:00:00 Immigrant      Vern~ B.C.
## 9 Betty Stills   bettys5~ 1997-09-06 00:00:00 Non-permanent r~ Cobb~ B.C.
## 10 Enrique Iglasias ei123@h~ 1985-10-16 00:00:00 Non-permanent r~ Cobb~ B.C.
## # i 10 more variables: orchard_id <chr>, owner_id <chr>, username_id <chr>,
## #   sns_worked <dbl>, sat_hrs <dbl>, trt_workers <dbl>, trt_manager <dbl>,
## #   cmf_manager <dbl>, sat_work_overall <dbl>, comments <chr>
```

2.2 Direct identifiers

These variables point to a specific person and will be removed in Part 3:

Variable	Why it is risky
worker_id	Full name (300 unique values)

Variable	Why it is risky
email_id	Email address (300 unique values)
username_id	Social-media handle (265 unique values)
owner_id	Orchard owner name

2.3 Dates

The column **age** stores **date of birth**, not a numeric age. Exact DOB is a strong quasi-identifier when combined with location and employer.

```
data.frame(
  earliest = as.character(min(as.Date(raw$age))),
  latest   = as.character(max(as.Date(raw$age))),
  unique_dates = length(unique(raw$age))
)
```

```
##      earliest      latest unique_dates
## 1 1970-02-20 2000-11-11           295
```

Exact DOB will be replaced with **5-year age bands** in Part 3.

2.4 Geography

city has 12 levels; province is always B.C. Small groups are easier to re-identify.

```
raw |>
  count(city, sort = TRUE) |>
  mutate(rare = n < K_THRESHOLD)
```

```
## # A tibble: 12 x 3
##   city      n rare
##   <chr>    <int> <lgl>
## 1 Victoria    33 FALSE
## 2 Summerland  32 FALSE
## 3 Keremeos    31 FALSE
## 4 Kelowna     29 FALSE
## 5 West Kelowna 29 FALSE
## 6 Peachland   27 FALSE
## 7 Cobble Hill 26 FALSE
## 8 Vernon      24 FALSE
## 9 Naramata    18 FALSE
## 10 Oliver     18 FALSE
## 11 Osoyoos    17 FALSE
## 12 Penticton  16 FALSE
```

```
rare_cities <- raw |>
  count(city) |>
  filter(n < K_THRESHOLD) |>
  pull(city)

cat("Rare cities (n < ", K_THRESHOLD, "):\n")
```

```
## Rare cities (n < 5 ):
```

```
rare_cities
```

```
## character(0)
```

```
raw |> count(immigration_stat, sort = TRUE)
```

```
## # A tibble: 3 x 2
##   immigration_stat      n
##   <chr>             <int>
## 1 Non-permanent resident 107
## 2 Non-immigrant         103
## 3 Immigrant             90
```

Note: Pool rare cities to "Other BC community", then replace all city names with codes (City_01, ...). Immigration counts are checked here; recoding is only needed if any group falls below 5.

2.5 External sources (free text)

The `comments` field can name people, employers, platforms, and birth years — even after other columns are cleaned.

```
raw |>
  filter(comments != "None") |>
  slice(1:10) |>
  select(comments)
```

```
## # A tibble: 10 x 1
##   comments
##   <chr>
## 1 I get that I'm a farm worker and grinding is part of the job, but it'd be ni-
## 2 I don't mind the actual work, but the people I work with and the owners like-
## 3 I only like this job because I can make good money, everything else sucks. I-
## 4 The hours and work are hard, but it's good money and I love the team we have-
## 5 Hard but great work
## 6 I'm an apple picker, it is what it is
## 7 It's a good summer job to help me pay for school, definitely not something I-
## 8 The owner's wife is such a sweetheart, but I don't like the owner himself. ~
## 9 I hate this job and everybody who works here. the guys I work with are awfu-
## 10 I don't love the job but I stay for the money and the routine.
```

Note: Remove `comments` entirely. Free text cannot be reliably de-identified without manual review.

2.6 Quasi-identifiers (orchards)

Orchard names link respondents to a specific workplace. They will be pseudonymized in Part 3.

```
raw |> count(orchard_id, sort = TRUE)
```

```
## # A tibble: 12 x 2
##   orchard_id      n
##   <chr>         <int>
## 1 Billie's Apples 33
## 2 Okanagan Fresh 32
## 3 Valley View Apples 31
## 4 Applejacks      29
## 5 West Kelowna Apples 29
## 6 Lakeside Groves 27
## 7 Country Apples  26
## 8 Bon Appetit     24
```

```
## 9 Golden Valley Farms      18
## 10 Ridgeline Orchard       18
## 11 Peak Harvest Co        17
## 12 Sunrise Orchards       16
```

Note: Replace orchard names with non-descriptive codes (Orchard_01, ...).

Part 3 — Transform

Apply each rule from Part 2. Run chunks in order; each step shows **10 rows** of the updated dataframe.

3.1 Remove direct identifiers and comments

```
direct_ids <- c("worker_id", "email_id", "username_id", "owner_id")

step1 <- raw |>
  select(-all_of(direct_ids), -comments)

cat("Step 3.1:", nrow(step1), "rows ×", ncol(step1), "columns\n")

## Step 3.1: 300 rows × 11 columns

head(step1, 10)

## # A tibble: 10 × 11
##   age                immigration_stat    city province orchard_id sns_worked
##   <dtm>              <chr>              <chr> <chr>    <chr>      <dbl>
## 1 1998-09-10 00:00:00 Non-immigrant      Kelo~ B.C.    Applejacks      2
## 2 1994-08-04 00:00:00 Immigrant          Kelo~ B.C.    Applejacks      4
## 3 1995-01-02 00:00:00 Non-permanent resid~ West~ B.C.    West Kelo~      1
## 4 1991-03-22 00:00:00 Non-permanent resid~ West~ B.C.    West Kelo~      2
## 5 1990-07-15 00:00:00 Non-permanent resid~ Vict~ B.C.    Billie's ~      2
## 6 1988-06-06 00:00:00 Immigrant          Vict~ B.C.    Billie's ~      5
## 7 2000-11-11 00:00:00 Non-immigrant      Vern~ B.C.    Bon Apple~      3
## 8 1992-12-18 00:00:00 Immigrant          Vern~ B.C.    Bon Apple~      1
## 9 1997-09-06 00:00:00 Non-permanent resid~ Cobb~ B.C.    Country A~      3
## 10 1985-10-16 00:00:00 Non-permanent resid~ Cobb~ B.C.    Country A~      6
## # i 5 more variables: sat_hrs <dbl>, trt_workers <dbl>, trt_manager <dbl>,
## #   cmf_manager <dbl>, sat_work_overall <dbl>
```

3.2 Generalize date of birth

```
to_age_band <- function(dob) {
  yrs <- as.integer(difftime(Sys.Date(), as.Date(dob), units = "days") / 365.25)
  if (yrs < 25)      "18-24"
  else if (yrs < 35) "25-34"
  else if (yrs < 45) "35-44"
  else if (yrs < 55) "45-54"
  else              "55+"
}

step2 <- step1 |>
  mutate(age_band = sapply(age, to_age_band)) |>
```

```

select(-age) |>
relocate(age_band, .before = immigration_stat)

cat("Step 3.2:", nrow(step2), "rows ×", ncol(step2), "columns\n")

## Step 3.2: 300 rows × 11 columns

head(step2, 10)

## # A tibble: 10 × 11
##   age_band immigration_stat city province orchard_id sns_worked sat_hrs
##   <chr>      <chr>      <chr> <chr>    <chr>      <dbl>    <dbl>
## 1 25-34      Non-immigrant Kelow~ B.C.    Applejacks      2      3
## 2 25-34      Immigrant      Kelow~ B.C.    Applejacks      4      2
## 3 25-34      Non-permanent resident West ~ B.C.    West Kelo~      1      1
## 4 35-44      Non-permanent resident West ~ B.C.    West Kelo~      2      3
## 5 35-44      Non-permanent resident Victo~ B.C.    Billie's ~      2      2
## 6 35-44      Immigrant      Victo~ B.C.    Billie's ~      5      3
## 7 25-34      Non-immigrant      Vernon B.C.    Bon Apple~      3      3
## 8 25-34      Immigrant      Vernon B.C.    Bon Apple~      1      4
## 9 25-34      Non-permanent resident Cobbl~ B.C.    Country A~      3      3
## 10 35-44      Non-permanent resident Cobbl~ B.C.    Country A~      6      1
## # i 4 more variables: trt_workers <dbl>, trt_manager <dbl>, cmf_manager <dbl>,
## #   sat_work_overall <dbl>

```

3.3 Suppress rare cities

```

step3 <- step2 |>
mutate(city = if_else(city %in% rare_cities, "Other BC community", city))

cat("Step 3.3:", nrow(step3), "rows ×", ncol(step3), "columns\n")

## Step 3.3: 300 rows × 11 columns

head(step3, 10)

## # A tibble: 10 × 11
##   age_band immigration_stat city province orchard_id sns_worked sat_hrs
##   <chr>      <chr>      <chr> <chr>    <chr>      <dbl>    <dbl>
## 1 25-34      Non-immigrant Kelow~ B.C.    Applejacks      2      3
## 2 25-34      Immigrant      Kelow~ B.C.    Applejacks      4      2
## 3 25-34      Non-permanent resident West ~ B.C.    West Kelo~      1      1
## 4 35-44      Non-permanent resident West ~ B.C.    West Kelo~      2      3
## 5 35-44      Non-permanent resident Victo~ B.C.    Billie's ~      2      2
## 6 35-44      Immigrant      Victo~ B.C.    Billie's ~      5      3
## 7 25-34      Non-immigrant      Vernon B.C.    Bon Apple~      3      3
## 8 25-34      Immigrant      Vernon B.C.    Bon Apple~      1      4
## 9 25-34      Non-permanent resident Cobbl~ B.C.    Country A~      3      3
## 10 35-44      Non-permanent resident Cobbl~ B.C.    Country A~      6      1
## # i 4 more variables: trt_workers <dbl>, trt_manager <dbl>, cmf_manager <dbl>,
## #   sat_work_overall <dbl>

step3 |> count(city, sort = TRUE)

## # A tibble: 12 × 2
##   city          n

```

```
##      <chr>      <int>
##  1 Victoria      33
##  2 Summerland    32
##  3 Keremeos      31
##  4 Kelowna       29
##  5 West Kelowna  29
##  6 Peachland     27
##  7 Cobble Hill   26
##  8 Vernon        24
##  9 Naramata      18
## 10 Oliver        18
## 11 Osoyoos       17
## 12 Penticton     16
```

3.4 Anonymize city

```
step4 <- step3 |>
  mutate(city = paste0(
    "City_", sprintf("%02d", as.integer(factor(city)))
  ))

cat("Step 3.4:", nrow(step4), "rows x", ncol(step4), "columns\n")
```

```
## Step 3.4: 300 rows x 11 columns
```

```
head(step4, 10)
```

```
## # A tibble: 10 x 11
##   age_band immigration_stat      city province orchard_id sns_worked sat_hrs
##   <chr>      <chr>          <chr> <chr>      <chr>      <dbl>   <dbl>
##  1 25-34      Non-immigrant      City_~ B.C.      Applejacks      2       3
##  2 25-34      Immigrant          City_~ B.C.      Applejacks      4       2
##  3 25-34      Non-permanent resident City_~ B.C.      West Kelo~      1       1
##  4 35-44      Non-permanent resident City_~ B.C.      West Kelo~      2       3
##  5 35-44      Non-permanent resident City_~ B.C.      Billie's ~      2       2
##  6 35-44      Immigrant          City_~ B.C.      Billie's ~      5       3
##  7 25-34      Non-immigrant      City_~ B.C.      Bon Apple~      3       3
##  8 25-34      Immigrant          City_~ B.C.      Bon Apple~      1       4
##  9 25-34      Non-permanent resident City_~ B.C.      Country A~      3       3
## 10 35-44      Non-permanent resident City_~ B.C.      Country A~      6       1
## # i 4 more variables: trt_workers <dbl>, trt_manager <dbl>, cmf_manager <dbl>,
## #   sat_work_overall <dbl>
```

```
step4 |> count(city, sort = TRUE)
```

```
## # A tibble: 12 x 2
##   city      n
##   <chr>   <int>
##  1 City_11    33
##  2 City_09    32
##  3 City_03    31
##  4 City_02    29
##  5 City_12    29
##  6 City_07    27
##  7 City_01    26
```



```
## 8 City_10      24
## 9 City_04      18
## 10 City_05     18
## 11 City_06     17
## 12 City_08     16
```

3.5 Pseudonymize orchard

```
step5 <- step4 |>
  mutate(orchard_id = paste0(
    "Orchard_", sprintf("%02d", as.integer(factor(orchard_id)))
  ))

deid <- step5

cat("Step 3.5 - final:", nrow(deid), "rows ×", ncol(deid), "columns\n")
```

```
## Step 3.5 - final: 300 rows × 11 columns
```

```
head(deid, 10)
```

```
## # A tibble: 10 x 11
##   age_band immigration_stat      city province orchard_id sns_worked sat_hrs
##   <chr>      <chr>           <chr> <chr>    <chr>      <dbl>   <dbl>
## 1 25-34      Non-immigrant      City_~ B.C.    Orchard_01      2       3
## 2 25-34      Immigrant          City_~ B.C.    Orchard_01      4       2
## 3 25-34      Non-permanent resident City_~ B.C.    Orchard_12      1       1
## 4 35-44      Non-permanent resident City_~ B.C.    Orchard_12      2       3
## 5 35-44      Non-permanent resident City_~ B.C.    Orchard_02      2       2
## 6 35-44      Immigrant          City_~ B.C.    Orchard_02      5       3
## 7 25-34      Non-immigrant      City_~ B.C.    Orchard_03      3       3
## 8 25-34      Immigrant          City_~ B.C.    Orchard_03      1       4
## 9 25-34      Non-permanent resident City_~ B.C.    Orchard_04      3       3
## 10 35-44      Non-permanent resident City_~ B.C.    Orchard_04      6       1
## # i 4 more variables: trt_workers <dbl>, trt_manager <dbl>, cmf_manager <dbl>,
## #   sat_work_overall <dbl>
```

```
deid |> count(orchard_id, sort = TRUE)
```

```
## # A tibble: 12 x 2
##   orchard_id      n
##   <chr>      <int>
## 1 Orchard_02     33
## 2 Orchard_07     32
## 3 Orchard_11     31
## 4 Orchard_01     29
## 5 Orchard_12     29
## 6 Orchard_06     27
## 7 Orchard_04     26
## 8 Orchard_03     24
## 9 Orchard_05     18
## 10 Orchard_09     18
## 11 Orchard_08     17
## 12 Orchard_10     16
```

What changed

Removed	Transformed
worker_id, email_id, username_id, owner_id, comments, exact DOB (age)	age → age_band; rare cities pooled; city → City_01 ...; orchard_id → Orchard_01 ...

Part 4 — Verify and Deliver

4.1 QA checks

Confirm the de-identified data is safe to export. All checks should show TRUE.

```
data.frame(  
  check = c(  
    "No direct ID columns",  
    "No comments column",  
    "No exact DOB column",  
    "Row count unchanged",  
    "Rare cities suppressed",  
    "Cities anonymized",  
    "Orchards pseudonymized"  
  ),  
  passed = c(  
    !any(direct_ids %in% names(deid)),  
    !"comments" %in% names(deid),  
    !"age" %in% names(deid),  
    nrow(deid) == nrow(raw),  
    !any(rare_cities %in% deid$city),  
    !any(unique(raw$city) %in% deid$city),  
    !any(unique(raw$orchard_id) %in% deid$orchard_id)  
  )  
)
```

```
##               check passed  
## 1   No direct ID columns   TRUE  
## 2     No comments column   TRUE  
## 3     No exact DOB column   TRUE  
## 4   Row count unchanged    TRUE  
## 5 Rare cities suppressed    TRUE  
## 6     Cities anonymized    TRUE  
## 7 Orchards pseudonymized   TRUE
```

```
data.frame(  
  rows      = nrow(deid),  
  columns    = ncol(deid),  
  smallest_city = min(table(deid$city)),  
  smallest_age_band = min(table(deid$age_band)),  
  smallest_orchard = min(table(deid$orchard_id))  
)
```

```
##   rows columns smallest_city smallest_age_band smallest_orchard  
## 1   300     11           16           13           16
```

Note: Each group should ideally have at least 5 records, or be pooled into a broader category.

4.2 Export de-identified dataset

This is the file you can share for analysis. It does **not** include the data key.

```
write_xlsx(deid, OUTPUT_FILE)

cat("Exported:", OUTPUT_FILE, "\n")

## Exported: WorkerSatisfaction_300rows_deidentified.xlsx

cat("Rows:", nrow(deid), "| Columns:", ncol(deid), "\n\n")

## Rows: 300 | Columns: 11

cat("Preview of exported data:\n")

## Preview of exported data:

head(deid, 10)

## # A tibble: 10 x 11
##   age_band immigration_stat      city province orchard_id sns_worked sat_hrs
##   <chr>      <chr>          <chr> <chr>      <chr>      <dbl>   <dbl>
## 1 25-34      Non-immigrant      City_~ B.C.      Orchard_01      2       3
## 2 25-34      Immigrant            City_~ B.C.      Orchard_01      4       2
## 3 25-34      Non-permanent resident City_~ B.C.      Orchard_12      1       1
## 4 35-44      Non-permanent resident City_~ B.C.      Orchard_12      2       3
## 5 35-44      Non-permanent resident City_~ B.C.      Orchard_02      2       2
## 6 35-44      Immigrant            City_~ B.C.      Orchard_02      5       3
## 7 25-34      Non-immigrant      City_~ B.C.      Orchard_03      3       3
## 8 25-34      Immigrant            City_~ B.C.      Orchard_03      1       4
## 9 25-34      Non-permanent resident City_~ B.C.      Orchard_04      3       3
## 10 35-44      Non-permanent resident City_~ B.C.      Orchard_04      6       1
## # i 4 more variables: trt_workers <dbl>, trt_manager <dbl>, cmf_manager <dbl>,
## #   sat_work_overall <dbl>
```

4.3 Data key file (DUMMY — illustrative only)

Keeping the data key file well protected is essential because it is the sole mechanism that can re-identify individuals in otherwise de-identified research data, and its compromise can undermine confidentiality protections and cause harm to participants. In the UBC environment, the key file must be stored separately from research data on UBC-approved secure systems, with access restricted to authorized personnel only and never stored on personal devices or unapproved cloud services.

Here is an example of a data key file for the work described above; this is a **dummy file provided for illustrative purposes only**, and in real-world research settings, such files must never be shared on public systems.

File path: data_key_file/WorkerSatisfaction_data_key_DUMMY.xlsx

Sheet	Contents
README	Dummy disclaimer and UBC storage requirements
orchard_key	Original orchard name → pseudonym code
city_key	City label after suppression → pseudonym code
rare_city_suppress	Cities pooled to "Other BC community"

The key holds **orchard and city code mappings only** — not worker names, emails, or other direct identifiers removed in step 3.1.

```
orchard_key <- step1 |>
  distinct(orchard_id) |>
  arrange(orchard_id) |>
  transmute(
    original_orchard = orchard_id,
    pseudonym_code   = paste0("Orchard_", sprintf("%02d", row_number()))
  )

city_key <- step3 |>
  distinct(city) |>
  arrange(city) |>
  transmute(
    city_after_suppression = city,
    pseudonym_code         = paste0("City_", sprintf("%02d", row_number()))
  )

if (length(rare_cities) > 0) {
  rare_city_key <- data.frame(
    original_city = rare_cities,
    pooled_to     = "Other BC community"
  )
} else {
  rare_city_key <- data.frame(
    original_city = "(none - all cities at or above k threshold)",
    pooled_to     = NA_character_
  )
}

key_readme <- data.frame(
  field = c(
    "File status",
    "Purpose",
    "Contains",
    "Storage requirement (UBC)"
  ),
  description = c(
    "DUMMY - illustrative example only; not for production use",
    "Reverse pseudonym codes for orchard and city in the de-identified file",
    "Orchard map, city map, rare-city suppression list - no worker names or emails",
    "Store separately from de-identified data on UBC-approved secure systems; authorized access only"
  )
)

cat("Orchard key:\n")
```

Orchard key:

```
head(orchard_key, 10)
```

```
## # A tibble: 10 x 2
##   original_orchard pseudonym_code
##   <chr>           <chr>
```

```
## 1 Applejacks Orchard_01
## 2 Billie's Apples Orchard_02
## 3 Bon Appletit Orchard_03
## 4 Country Apples Orchard_04
## 5 Golden Valley Farms Orchard_05
## 6 Lakeside Groves Orchard_06
## 7 Okanagan Fresh Orchard_07
## 8 Peak Harvest Co Orchard_08
## 9 Ridgeline Orchard Orchard_09
## 10 Sunrise Orchards Orchard_10
```

```
cat("\nCity key:\n")
```

```
##
## City key:
```

```
city_key
```

```
## # A tibble: 12 x 2
##   city_after_suppression pseudonym_code
##   <chr>                  <chr>
## 1 Cobble Hill           City_01
## 2 Kelowna               City_02
## 3 Keremeos              City_03
## 4 Naramata              City_04
## 5 Oliver                City_05
## 6 Osoyoos               City_06
## 7 Peachland             City_07
## 8 Penticton             City_08
## 9 Summerland            City_09
## 10 Vernon               City_10
## 11 Victoria             City_11
## 12 West Kelowna         City_12
```

```
dir.create(KEY_DIR, showWarnings = FALSE)
```

```
write_xlsx(
  list(
    README = key_readme,
    orchard_key = orchard_key,
    city_key = city_key,
    rare_city_suppress = rare_city_key
  ),
  KEY_FILE
)
```

```
cat("Data key exported:", KEY_FILE, "\n")
```

```
## Data key exported: data_key_file/WorkerSatisfaction_data_key_DUMMY.xlsx
```

```
cat("Sheets: README, orchard_key, city_key, rare_city_suppress\n")
```

```
## Sheets: README, orchard_key, city_key, rare_city_suppress
```

```
cat("Store separately from", OUTPUT_FILE, "- never on public systems.\n")
```

```
## Store separately from WorkerSatisfaction_300rows_deidentified.xlsx - never on public systems.
```

Closing Summary

Stage	What changed
Direct identifiers	Removed <code>worker_id</code> , <code>email_id</code> , <code>username_id</code> , <code>owner_id</code>
External sources	Removed <code>comments</code>
Dates	<code>age</code> (DOB) → <code>age_band</code> (5-year groups)
Geography	Rare cities pooled; all cities → <code>City_01</code> ...
Quasi-identifiers	<code>orchard_id</code> → <code>Orchard_01</code> ...; mappings in <code>data_key_file/WorkerSatisfaction_data_key_DUMMY.xlsx</code> (dummy, separate storage)

The de-identified dataset keeps all satisfaction and treatment ratings for analysis while substantially lowering re-identification risk. The data key remains the only link back to original orchard and city names — protect it accordingly.